OVERVIEW

Policy focus:	Dengue surveillance, early-warning signals, and accountable local response
Evidence base:	SIGMA City Evidence and Comparison Matrix, updated 19 October 2026; cited LGU records; WHO technical guidance; Republic Act No. 7160

Executive Summary

SIGMA is intended to support local public-health planning by evaluating whether statistical early-warning methods can identify increases in dengue cases earlier or more usefully than conventional surveillance thresholds. Its proposed approach compares the Early Aberration Reporting System (EARS), particularly C1, C2, and C3, with the Department of Health (DOH) and Philippine Integrated Disease Surveillance and Response (PIDSR) threshold methodology.

The reviewed evidence matrix suggests that the primary policy opportunity is to connect surveillance signals to a documented, accountable, and evaluable response process.

The four LGUs represented in the workbook demonstrate different starting points:

I. **Quezon City**
- ➢ Documented dengue response activities. A network of 66 health centers in its 2024–2025 annual report, and reported outbreak-response outcomes. However, the reviewed records do not establish a complete numeric alert-to-response protocol or independently demonstrate the causal effect of the interventions.

II. **Pasig City**
- ➢ Reported a broad prevention package, including community cleanups, vector-control activities, ovitraps, larvicide placement, health education, and hospital coordination. The reviewed release does not provide a reproducible case series or quantify the reported decline.

III. **Davao City**
- ➢ Clearest formal policy architecture in the reviewed evidence: Ordinance No. 0401-20, series of 2020, implementing rules under Executive Order No. 15, series of 2023, and a barangay task-force structure. However, a SIGMA-compatible alert threshold, response-time standard, and linked action log were not established in the reviewed sources.

IV. **Bacolod City**
- ➢ Official resolutions-index entry for City Resolution 662, described as directing intensified dengue and leptospirosis action. The full resolution and evidence of implementation remain unverified.

Across these cases, the matrix repeatedly identifies gaps in documented alert triggers, response-time requirements, surveillance completeness, budget reconciliation, action records, and independent evaluation.

Recommended policy direction

SIGMA should be developed and piloted under a Dengue Signal-to-Action Framework comprising five essential components:

1. Validated surveillance inputs. Verify data completeness, reporting intervals, geographic consistency, case definitions, and data freshness before interpreting signals.
2. Interpretable statistical warnings. Validate EARS outputs through chronological backtesting against a clearly defined DOH/PIDSR reference method, assessing detection accuracy, false alarms, missed signals, and warning lead time.
3. Human-led epidemiological review. Require an authorized public-health officer to assess each signal before it informs operational decisions.
4. Documented response. Record the action taken, responsible office, response time, completion status, resources used, and grounds for closure.
5. Continuous evaluation. Assess predictive performance, operational timeliness, implementation, equity, expenditure, and health outcomes separately.

SIGMA's authority and limitations must be expressly defined. An EARS signal shall not, by itself, constitute an official outbreak declaration, determine clinical decisions, mandate fogging, or justify penalties. SIGMA shall complement existing public-health surveillance systems and remain subject to applicable DOH guidelines, established protocols, and the authority of responsible health officials. Pilot expansion should be contingent on documented evidence of its effectiveness, operational feasibility, and accountability.

1. Policy Background and Problem Definition

1.1 The public-health problem

Dengue prevention and control require surveillance, early recognition of possible increases, appropriate field investigation, community participation, clinical preparedness, and coordinated local action.

Conventional thresholds provide interpretable reference points for public-health surveillance. However, a statistical early-warning method may identify unusual increases that warrant investigation before conventional thresholds are crossed. Whether EARS provides a meaningful advantage in the Philippine context is an empirical question that must be evaluated using valid historical data.

SIGMA's retrospective comparison of EARS C1, C2 and C3 against DOH/PIDSR alert and epidemic thresholds. The proposed evaluation includes sensitivity, precision, false-alert rate, missed threshold exceedances and warning lead time.

These are appropriate categories of evaluation, but the proposal's intended analysis should not be confused with completed validation. No predictive-performance results are established by the Excel evidence matrix alone.

1.2 The operational policy gap

The city evidence indicates that dengue programs and activities already exist. The more specific gap concerns how a surveillance signal becomes an accountable response.

A complete process should answer:

1. What data generated the signal?
2. Was the reporting series complete and geographically valid?
3. Which officer receives the signal?
4. Who verifies its epidemiological significance?
5. What determines whether the signal is escalated?
6. Which actions are authorized and appropriate?
7. When must the assigned action begin and finish?
8. What evidence confirms that the action was completed?
9. How are the response, expenditure and subsequent case trends reviewed?

Without these elements, a dashboard may display a statistical anomaly without establishing whether the information was reviewed, acted upon or useful.

1.3 Proposed policy objective

To establish a transparent, evidence-led and locally accountable dengue signal-to-action system that supports earlier review of potential increases, strengthens coordination between local health offices and barangays, documents prevention and response activities, and enables evaluation of timeliness, resource use, implementation and outcomes.

1.4 Specific objectives

1. Improve the validity and traceability of data used for dengue early-warning analysis.
2. Evaluate EARS performance against an appropriate DOH/PIDSR reference method.

3. Establish clear responsibilities for signal review, escalation, action and closure.
4. Link surveillance signals to documented activities, resource use and follow-up.
5. Improve LGU capacity to assess program performance without overstating causal conclusions.
6. Provide a replicable policy framework that can be adapted to local institutional arrangements.

2. Evidence Base and Methodological Boundaries

2.1 Scope of the Excel evidence matrix

The workbook is a first evidence-backed comparative scan, not an exhaustive assessment of Philippine LGUs.

Its deeper evidence covers Quezon City, Pasig City and Davao City, with a preliminary Bacolod City lead. Manila, Cebu, Iloilo and Cagayan de Oro are identified as possible future research sites, but the workbook does not establish equivalent evidence coverage for these cities.

2.2 Evidence classifications

The analysis follows five distinctions.

Documented policy adoption: A law, ordinance, executive order, resolution or approved policy instrument establishes a formal mandate.

Reported implementation: An official source describes activities, organizational structures or services. This does not necessarily establish complete coverage or consistent implementation.

Reported outcome: A government source reports a change in cases, deaths or other indicators. Unless supported by a suitable evaluation, this should not be interpreted as proof that a particular intervention caused the change.

Budget or procurement evidence: A record establishes a particular stage of financing or procurement. A planned amount is not the same as a contract award, delivered goods, disbursement or actual expenditure.

Unknown or partially verified: The reviewed sources do not establish the relevant claim, or only an index entry or indirect reference is available. This does not prove that no additional evidence exists.

3. Technical Assessment of the Dengue Dataset

3.1 Dataset description

The SIGMA master proposal and Excel workbook describe a historical dengue dataset obtained through the Humanitarian Data Exchange (HDX), attributed to the DOH Epidemiology Bureau.

The proposal-reported characteristics are:

Dimension	Description
Record count	32,701 records
Geographic coverage	126 locations across 17 regions
Period	10 January 2016-10 January 2021
Temporal quality	Values are reported
Geographic matching	City-name joins are subject to validation

These characteristics are reported in the SIGMA source material and must be checked against the actual raw dataset before publication or operational use.

3.2 Required data-quality controls

Before running the algorithms, SIGMA should:

1. Preserve the raw source file and record its download date, source URL, version and checksum where practical.
2. Document every variable, including its definition, unit, reporting period and allowed values.
3. Check duplicate records, invalid dates, missing fields and implausible values.
4. Establish the expected reporting frequency and identify missing or irregular periods.
5. Distinguish an observed zero from an absent or unreported observation.
6. Validate location names, geographic levels and region labels against the raw source.
7. Document any exclusions, corrections, imputation or aggregation.
8. Version the cleaned dataset and transformation code.
9. Reconcile case definitions, geographic coverage and reporting periods before combining different datasets.
10. Display a data-quality status alongside any generated signal.

The dataset should not be silently converted into a complete weekly series if its reporting intervals are irregular. Any transformation must be documented and justified.

3.3 Geographic validation

The workbook's LGU Name Check sheet marks Quezon City, Pasig City, Davao City and Bacolod City as requiring review against the raw location list.

Therefore, the city-policy evidence and historical case records must not yet be treated as Historical geographic classifications should be preserved in the analytical record. A present-day regional label should not be substituted into historical observations without documenting the crosswalk.

3.4 Recommended EARS evaluation

SIGMA should evaluate C1, C2 and C3 against a clearly specified reference method through chronological backtesting.

The evaluation protocol should define, before results are examined:

1. The event or threshold being predicted
2. The reference method and version
3. Baseline and comparison windows
4. Signal-generation parameters
5. Geographic inclusion rules
6. Handling of missing and delayed reports
7. The period between a signal and the event that counts as useful warning
8. The method for handling multiple signals associated with one event

Recommended metrics include:

Metric	Purpose
Sensitivity	Measures the share of reference events detected by the method.
Precision	Measures the share of signals followed by the defined reference event within the evaluation window.
False-alert rate	Quantifies signals not followed by the defined reference event, using an explicitly stated denominator.

Missed threshold exceedances	Identifies reference events that were not signalled in time.
Warning lead time	Measures the interval between a signal and the defined reference event.
Reporting completeness	Measures whether expected observations were received.
Reporting delay	Measures how late the information becomes available for review.

Use chronological training and evaluation periods rather than randomly mixing past and future observations. Where feasible, test whether results remain stable across time periods and geographic units.

A threshold exceedance is not necessarily an independently confirmed outbreak. SIGMA must define its reference outcome precisely and report the limitations of using it as a proxy for an outbreak.

Technical recommendation: EARS should remain a decision-support method until its performance, data requirements and operational value have been evaluated and reviewed by responsible public-health officers.

WHO's Operational Guide: The Early Warning and Response System (EWARS) for Dengue Outbreaks and Technical Handbook for Dengue Surveillance, Outbreak Prediction/Detection and Outbreak Response provide relevant technical guidance. They do not, by themselves, validate SIGMA's implementation or establish its predictive accuracy.

4. Comparative LGU Analysis

4.1 Quezon City

Evidence documented in the workbook

The Quezon City Government Annual Report for July 2024–June 2025 describes dengue response activities beginning in February 2025, including:

- Intensified monitoring in high-risk areas
- Awareness and supplemental campaigns

- Daily free dengue testing and consultations at 66 health centers
- Targeted cleanup drives
- Fogging, spraying and larviciding where cases were detected

The report states that dengue cases had declined by 90% by April 2025 and that 123 barangays had been cleared from outbreak status.

These are official city-reported outcomes. The workbook does not include the underlying weekly case series, the precise comparison window, clearance criteria, intervention coverage, linked costs or an independent causal evaluation.

Budget evidence

The FY2024 Annual Procurement Plan lists four identified QCHD vermin-abatement items:

Item	**Planned amount**
Diflubenzuron	PHP 750,000
Beta-cyfluthrin	PHP 650,000
Etofenprox	PHP 800,000
Clothianidin + Deltamethrin	PHP 1,560,000
Total of these four listed lines	**PHP 3,760,000**

This is a total of identified planned procurement amounts only. It is not the city's total dengue budget and does not establish contract awards, delivery, payment or actual use. The city's 2020 development/performance plan also identifies dengue morbidity and mortality targets. The workbook flags the need to reconcile the reported baseline of 19 cases and 39 deaths with citywide surveillance definitions before comparing it with other figures.

Analysis

Quezon City demonstrates an established service network and a broad set of reported response activities. The reviewed evidence does not, however, establish a complete numeric signal trigger, alert recipient, response-time requirement, reporting-completeness metric or linked activity-cost-outcome record.

Recommendation for SIGMA

SIGMA should support the existing health-center and barangay network rather than assume that the city lacks a dengue program. Priority tasks are to:

- Obtain the weekly case and death series for 2024–2026.

- Verify the reported 90% decline and the criteria for clearing barangays from outbreak status.
- Reconcile planned procurement with awards, delivery, disbursement, inventory and use.
- Establish a documented process linking validated signals to the responsible health office and barangay.
- Measure the timeliness, completion and coverage of response activities.

4.2 Pasig City

Evidence documented in the workbook

A Pasig City Government release dated 6 March 2025 describes a Dengue Task Force and a prevention package involving:

- Community cleanup drives
- Misting, spraying and fogging
- Fogging in public and private schools
- Thermal fogging in areas with case clustering
- Hospital coordination to strengthen fast lanes
- Health education by Pasig Health Aides
- Larvicide placement
- Ovitraps

The release describes a decline in cases but does not provide a reproducible case series, precise comparison dates, a denominator or independently verified outcome measures. The reviewed release does not establish the ordinance number or date of the formal instrument creating the task force.

Budget evidence

The workbook records two amounts—PHP 26,844,192.00 and PHP 26,233,312.00—in the relevant row of the city's FY2025 first-semester procurement monitoring report.

The original column headings, contract status, deliveries and payments must be verified before either amount is described as an award or expenditure. The scope must also distinguish dengue-specific costs from any shared sanitation allocation.

Analysis

Pasig's documented approach combines community prevention, vector-control activity, health education and clinical coordination. However, the reviewed evidence is descriptive

and does not establish a formal numeric alert-to-action standard or quantify the reported decline.

Recommendation for SIGMA

SIGMA should support cluster verification and consistent documentation of why an intervention was selected. It should also track:

- Case-series completeness and reporting delay
- Cluster verification time
- Assigned and completed actions
- Intervention coverage by location
- Hospital coordination indicators, if records are available
- Procurement and expenditure reconciliation

A reported decline should not be attributed to fogging, ovitraps, the task force or any single intervention without a suitable evaluation.

4.3 Davao City

Evidence documented in the workbook

The reviewed source register identifies:

- Ordinance No. 0401-20, series of 2020
- Executive Order No. 15, series of 2023, containing implementing rule
- An inter-agency task force
- Barangay Mosquito-Borne Diseases Task Force
- The city's 4S strategy: search and destroy breeding sites, self-protection, seek early consultation, and support indoor/outdoor spraying and fogging

A city communication urged barangays to establish task forces. Another source reported an enforcement mechanism with graduated penalties. The workbook notes that citation-ticket details were still being finalized at the time of the relevant report. The city reported 3,335 dengue cases in 2025 compared with 7,157 in 2024, a reported decrease of 53.54%. The workbook also records a separate Week 22 comparison of 1,682 cases versus 1,942 and 12 deaths versus 18. These figures are city-reported counts. They are not population-adjusted rates and do not establish that a particular program caused the decline. The reviewed sources do not establish a peso budget for the program, a SIGMA-compatible numeric signal threshold, a response-time service level or a complete trigger-to-action log.

Analysis

Davao has the clearest formal policy architecture among the four reviewed cases because a specific ordinance, implementing rules and barangay task-force structure are documented. The opportunity is to make that architecture measurable and auditable, rather than create a parallel structure that duplicates the existing task force.

Recommendation for SIGMA

1. Map the existing ordinance and implementing rules to actual operational responsibilities.
2. Obtain the approved program budget, procurement records and expenditure reports.
3. Obtain weekly barangay-level case data, deaths, population denominators where available, reporting completeness and task-force activity logs.
4. Connect reviewed signals to existing health-office and barangay responsibilities.
5. Measure task-force coverage, response timeliness and completion of assigned activities.
6. Verify the current status and actual operation of enforcement provisions before describing them as implemented.

4.4 Bacolod City

Evidence documented in the workbook

The official 2025 resolutions index lists City Resolution 662, described as directing the City Health Office and all barangays to intensify dengue and leptospirosis response.

The full resolution, exact legal status, operative requirements, implementation, budget and outcomes were not established in the reviewed source.

Analysis

Bacolod should be treated as a preliminary evidence lead rather than a confirmed example of successful implementation.

Recommendation for SIGMA

Obtain and review the full resolution, its legal status and date, implementation/accomplishment reports, budget and procurement records, weekly dengue data, and barangay activity logs before drawing conclusions about its policy model.

4.5 Comparative findings

Dimension	Evidence-based assessment	Recommended SIGMA response
Policy adoption	Davao has a cited ordinance and implementing rules; QC has program and performance-plan documents; Pasig has a task force referenced in a city release; Bacolod has an index listing only.	Maintain a register of full legal instruments, status, dates, mandates and implementing rules.
Implementation	Multiple activities are documented in QC, Pasig and Davao.	Track actual coverage and completion, not announcements alone.
Alert-to-response	The reviewed city sources did not establish a complete numeric trigger, named recipient, response deadline, and closure process.	Adopt a documented local signal-to-action protocol.
Surveillance quality	Reporting frequency, completeness and delay metrics are generally not supplied in the reviewed sources.	Establish common reporting-quality indicators.
Budget and resources	The workbook identifies planned procurement in QC, amounts requiring interpretation in Pasig, and no verified peso budget for Davao in the cited material.	Reconcile financial stages and link expenditure to activities.
Outcomes	QC and Davao report declines; Pasig reports a qualitative decline; Bacolod outcomes are not established.	Use standardized definitions, reproducible time windows and appropriate evaluation.
Evidence strength	Most sources are official records, but reported outcomes are not independent causal evaluations.	Display source type, verification status, date, limitations and causal status.

5. Proposed Dengue Signal-to-Action Policy

The following provisions are proposed for adaptation by participating LGUs. They are not represented as existing law. The final instrument should be reviewed by the local legal office, health authorities and relevant data-governance personnel.

5.1 Proposed policy statement

The LGU shall establish a Dengue Signal-to-Action Protocol to support timely, documented and proportionate public-health review of surveillance signals.

Where SIGMA is deployed, it shall operate as a decision-support system. A statistical signal shall not, by itself, constitute an official outbreak declaration, determine clinical care, authorize chemical application or trigger a penalty. Authorized public-health officers shall review relevant context and approve actions in accordance with applicable national guidance and local policy.

5.2 Proposed minimum policy provisions

Provision 1: Designated accountability. The City or Municipal Health Officer, or a formally designated surveillance lead, shall own epidemiological interpretation. Barangay health personnel and other agencies shall have assigned responsibilities under the approved local protocol.

Provision 2 : Data governance. The LGU shall identify approved data sources, reporting units, reporting frequency, case definitions, access roles, correction procedures, retention requirements and data-sharing arrangements.

Provision 3: Signal classification. The LGU shall distinguish routine monitoring, statistical aberration requiring review, corroborated local increase and officially declared alert or outbreak status. Labels and criteria must be documented and approved by responsible authorities.

Provision 4: Human review. A designated officer shall assess reporting completeness, missing or delayed reports, epidemiological context, available field information and applicable DOH/PIDSR procedures before escalating a signal.

Provision 5: Proportionate response. The LGU shall maintain an approved menu of response options. These may include data verification, case or cluster investigation,

intensified surveillance, risk communication, breeding-site inspection and removal, clinical readiness review and vector-control activities where technically justified.

Provision 6: Response timelines. The LGU shall set locally feasible service levels for acknowledgement, verification, decision, action initiation and closure. Targets must reflect local capacity and be monitored. SIGMA should not invent a universal response deadline unsupported by local policy or operational evidence.

Provision 7: Documentation. Each signal shall have a traceable record of its data source, algorithm and version, geographic unit, date, rationale, reviewer, decision, action owner, timestamps, completion evidence, resources used and closure reason.

Provision 8: Escalation. The protocol shall specify how unresolved signals, data-quality failures, threshold exceedances and events requiring assistance are referred to the appropriate health office or national authority.

Provision 9: Review and learning. The LGU shall periodically review signals, missed events, false alerts, incomplete actions, delayed reports and unintended effects.

Provision 10: Transparency and privacy. Aggregate performance should be reported where appropriate, while personal and sensitive health information is protected.

6. Response Design and Safeguards

SIGMA shall not treat every statistical signal as grounds for fogging or spraying. The appropriate response shall depend on data validity, corroborating evidence, local conditions, and applicable public-health guidelines. Accordingly, the following safeguards shall apply:

1. Incomplete or delayed reports. Verify reporting units and data pipelines before interpreting increases in reported cases as a genuine surge.
2. Uncorroborated statistical signals. Require epidemiological review and, where appropriate, field verification. No outbreak shall be declared solely on the basis of an EARS signal.
3. Corroborated epidemiological concerns. Initiate appropriate investigation, notify relevant health teams, strengthen targeted communication, and assess necessary preventive or clinical-readiness measures. Document the basis for each intervention.
4. Vector-control measures. Require compliance with approved technical criteria and implementation by qualified personnel. Record the authorization, location, timing, products used, quantities, safety measures, and completion status.
5. Official thresholds and escalation. Where applicable DOH/PIDSR thresholds are met or official escalation is required, follow established surveillance, notification, and response procedures. SIGMA shall not replace these processes.

6. Severe cases, deaths, or credible field alerts. Initiate review through existing public-health channels, regardless of whether SIGMA generates a signal. The absence of a model alert shall not delay urgent reporting or professional judgment.

For accountability, the protocol shall distinguish three classifications: a statistical signal, a reviewed epidemiological concern, and an officially declared outbreak. Each classification shall have defined criteria and corresponding procedures. A statistical signal shall not, by itself, establish an epidemiological finding or authorize a public-health intervention.

7. Monitoring and Evaluation Framework

SIGMA shall adopt a monitoring and evaluation framework that separately assesses data quality, model performance, response timeliness, implementation, financial accountability, health outcomes, and equity. The implementing agency shall define each indicator, its calculation method, reporting frequency, and responsible office before pilot implementation.

Indicator	Measurement	Frequency
Reporting completeness	Reports received ÷ reports expected	Weekly
Reporting delay	Time from reporting-period end to receipt and validation	Weekly
Geographic match rate	Records assigned validated locations ÷ records requiring assignment	Each data refresh
Signal review time	Time from signal generation to documented review	Per signal; monthly summary
Action initiation time	Time from validated decision to first documented action	Per action
Action completion rate	Actions completed by deadline ÷ actions due	Monthly
Signal confirmation proportion	Signals corroborated under predefined criteria ÷ signals reviewed	Quarterly
Sensitivity and precision	Model performance against predefined reference events	Backtesting and periodic review

False-alert rate	Signals not followed by the defined reference event, with denominator specified	Backtesting and periodic review
Warning lead time	Time between signal generation and the reference event or threshold crossing	Backtesting and periodic review
Intervention coverage	Sites or households reached ÷ target identified	Per activity
Budget traceability	Relevant financial records reconciled across procurement and payment stages	Quarterly
Dengue cases and deaths	Counts and, where valid, rates using consistent definitions and denominators	Weekly or monthly
Equity and coverage gaps	Differences in reporting, response times, and intervention coverage across locations	Quarterly

The implementing agency shall maintain supporting records sufficient to verify reported results and shall document material data limitations. Indicators shall be interpreted according to their defined measurement methods and shall not be treated as interchangeable measures of program effectiveness.

A decline in dengue cases following implementation shall not, by itself, establish that SIGMA caused the reduction. Evaluation shall account, where applicable, for seasonality, reporting changes, population differences, concurrent interventions, and other relevant factors.

Where feasible, the implementing agency should use a prespecified interrupted time-series or other appropriate quasi-experimental design to assess program effects. Where the assumptions necessary for causal inference are not satisfied, findings shall be reported as descriptive associations rather than causal conclusions.

8. Budget and Resource Accountability

The workbook illustrates why budget stages must be kept distinct. SIGMA should not label a planned procurement amount as actual spending or infer program effectiveness from expenditure alone.

8.1 Required financial stages

For each relevant program or activity, the LGU should maintain traceability across:

1. Approved appropriation and program budget.
2. Annual Investment Program and Annual Procurement Plan, where applicable.

3. Approved Budget for the Contract and procurement process.
4. Contract award or purchase order.
5. Obligation and delivery/inspection/acceptance.
6. Disbursement and payment.
7. Inventory issue, service delivery or documented activity.
8. Actual expenditure and, where feasible, unit cost.

8.2 City-specific follow-up

Quezon City: Reconcile the four identified FY2024 vermin-abatement lines totaling PHP 3.76 million against awards, purchase orders, delivery, inspection, payment, inventory and actual use. This is not the city's total dengue budget.

Pasig City: Verify the original column labels and status of the PHP 26,844,192.00 and PHP 26,233,312.00 amounts. Confirm whether the relevant allocation covers sanitation and dengue activities jointly, and reconcile the values with delivery and payment records.

Davao City: Obtain the approved budget, AIP, obligations, procurement and expenditure records associated with the Mosquito-Borne Diseases Prevention and Control Program.

Bacolod City: Obtain the full resolution, relevant AIP and budget records, procurement information, implementation reports and accomplishments before assessing resources.

8.3 Recommended SIGMA feature

Create an activity-level expenditure ledger with a unique activity identifier. This would allow a signal, approved action, responsible unit, resource requirement, expenditure record and completion evidence to be linked without implying that spending caused a particular health outcome.

9. Governance, Privacy and Safety

SIGMA should implement the following safeguards:

1. Assign named roles for data stewardship, epidemiological review, action ownership, supervision and system administration.
2. Use the minimum data necessary for policy analysis; prefer aggregate or de-identified records for dashboards.
3. Apply access controls to personal and sensitive health information.
4. Maintain an audit trail for data corrections, parameter changes, algorithm versions, user decisions and overrides.
5. Display when data are stale, incomplete, unvalidated or not comparable.

6. Allow authorized health officers to override model suggestions, with the reason documented.
7. Prevent automated outbreak declarations, clinical decisions, penalties or chemical applications based solely on a model output.
8. Publish aggregate methodology and performance indicators where appropriate without exposing identifiable health information.
9. Establish data-sharing and governance agreements before operational integration.

Republic Act No. 7160, the Local Government Code of 1991, provides a general statutory framework for local health service delivery and health-officer responsibilities. The final local policy should be reviewed against the statute and other applicable national requirements rather than treating SIGMA as an independent legal authority.

10. Implementation Roadmap

Implementation Phases and Indicative Timeline

The following phases constitute proposed planning estimates and shall not be construed as representations of current project status or guaranteed completion dates.

Phase	Indicative period	Required deliverables
0. Evidence and governance readiness	Weeks 1–4	Raw dataset and schema; source register; geographic validation; available LGU records; designated data owners and data-sharing arrangements.
1. Data quality and reproducibility	Weeks 5–8	Version-controlled cleaned data; data dictionary; missing-data and reporting-interval audit; documented transformations and limitations.
2. Backtesting and validation	Weeks 9–14	Predefined EARS comparison; chronological evaluation; performance metrics; error analysis and review.
3. Protocol co-design	Weeks 11–16, in parallel	LGU-approved signal classifications, response options, assigned responsibilities, deadlines, escalation procedures, closure criteria, and action logs.

4. Controlled pilot	Subject to readiness approval	Trained personnel; parallel testing or controlled deployment; periodic assessment of data quality, false alerts, and operational burden.
5. Evaluation and scale decision	After a predefined pilot period	Evaluation of model performance, response timeliness, action completion, resource use, equity, and health outcomes.

Progression to subsequent phases shall be subject to documented readiness assessments and completion of applicable prerequisites. Expansion shall be authorized only upon demonstrated data quality, practical utility, acceptable false-alert burden, operational feasibility, and adequate governance arrangements. The ability to generate dashboard signals alone shall not constitute sufficient justification for wider deployment.

11. Priority Actions by LGU

11. Priority Actions by LGU

SIGMA integration proceed according to each local government unit's (LGU) evidence gaps, existing institutional arrangements, and readiness for implementation. For Quezon City, priority should be given to obtaining weekly dengue cases and deaths for 2024–2026, defining the criteria underlying the reported 90% decline and clearance claims, and reconciling procurement records with budget execution. SIGMA should link validated signals to health-center and barangay action records. For Pasig City, the LGU should provide the weekly case and death data supporting the reported decline, clarify the legal basis of its task force, and verify procurement, contract, and payment records. Integration should document cluster verification, intervention rationale, and available clinical coordination indicators.

For Davao City, the immediate priorities are to obtain budget, Annual Investment Program (AIP), and expenditure records; verify ordinance implementation and enforcement status; and collect weekly barangay-level data and task-force activity logs. SIGMA should complement the existing task-force structure through documented signal review, response tracking, and performance monitoring. For Bacolod City, Resolution No. 662 should be retrieved and reviewed to establish its legal status, implementation, budget allocation, and documented outcomes. Until these matters are verified, Bacolod should remain a provisional comparison case. Meanwhile, Manila, Cebu, Iloilo, and Cagayan de Oro should be assessed using the same primary-source evidence standards before inclusion in comparative analysis or SIGMA integration.

Across all LGUs, inclusion and expansion should be contingent on a complete evidence register, validated geographic classifications, and verifiable implementation records. This

approach will strengthen comparability, prevent unsupported conclusions about local performance, and ensure that SIGMA integration is based on documented institutional readiness rather than the mere availability of reported data.

Final Recommendations

1. Adopt the signal-to-action protocol before operationally relying on SIGMA. A dashboard without named owners, deadlines and closure rules is not a complete early-warning system.
2. Validate the data before making geographic risk claims. The historical dataset's reporting intervals, missingness, lack of denominators, absence of a laboratory-confirmation field and unvalidated city joins materially limit interpretation.
3. Treat EARS as a method to evaluate, not a proven capability. Report sensitivity, precision, false-alert rate, missed events and lead time against pre-defined reference outcomes before making operational claims.
4. Use existing LGU structures as the implementation foundation. Davao's task-force architecture, Quezon City's health-center network and Pasig's community/clinical package are different starting points and should be adapted rather than replaced by a uniform model.
5. Prioritize operational traceability. Track the entire sequence from signal generation through review, decision, activity completion, resource use and follow-up.
6. Separate budget stages. Appropriation, planned procurement, award, obligation, delivery, disbursement and actual expenditure must be recorded distinctly.
7. Avoid unsupported causal claims. Reported year-to-year declines are important observations, but do not prove that one task force or intervention caused the decline.
8. Use traceable evidence in the policy module. Each city claim should identify the source, issuing body, publication date, reference period, evidence type, verification status and unresolved questions.
9. Require an evaluation before scaling. The pilot should demonstrate acceptable data quality, practical utility, manageable false-alert burden, action completion and governance readiness.

Limitations and Outstanding Verification

This document is based on the six-sheet Excel evidence matrix, its referenced sources and the stated SIGMA master-proposal dataset description. The following limitations remain:

1. The workbook is not an exhaustive census of Philippine LGUs.
2. The raw city-level case series was not contained in the evidence matrix; this document therefore does not calculate new city incidence rates or directly join policy records to case observations.
3. Government annual reports and news releases establish what an LGU reported; they do not automatically constitute independent implementation audits.
4. Bacolod's full Resolution 662 has not been reviewed in the evidence base.
5. The accounting interpretation of Pasig's procurement-monitoring amounts remains subject to verification.
6. Planned procurement values in Quezon City are not evidence of actual expenditure.
7. Davao's cited legal framework does not itself establish the peso budget or causal impact of the program
8. Proposed signal labels, response-time targets and policy provisions require local public-health, legal, budget and data-governance review before adoption.

References

R1. World Health Organization. (2017). Operational Guide: The Early Warning and Response System (EWARS) for Dengue Outbreaks.
https://www.who.int/publications/i/item/9789241512053
Technical guidance for organizing, analyzing and interpreting surveillance data for dengue early warning and response.

R2. World Health Organization. (2016). Technical Handbook for Dengue Surveillance, Outbreak Prediction/Detection and Outbreak Response.
https://www.who.int/publications/i/item/9789241549738
Technical reference for dengue surveillance, outbreak prediction/detection and response.

R3. Republic of the Philippines. Republic Act No. 7160 — Local Government Code of 1991. Official Gazette.
https://officialgazette.gov.ph/1991/10/10/republic-act-no-7160/
Statutory framework for local government functions and local health-officer responsibilities.

R4. Humanitarian Data Exchange. Philippine Dengue Cases and Deaths, dataset attributed in SIGMA's master proposal to the DOH Epidemiology Bureau.
https://data.humdata.org/dataset/philippine-dengue-cases-and-deaths
The actual downloaded file version, metadata and coverage must be revalidated before analysis.

R5. Quezon City Government. (2025). Local Government Annual Report July 2024–June 2025.

https://quezoncity.gov.ph/wp-content/uploads/2025/12/Quezon-City-Local-Government-Annual-Report-2024-2025.pdf
Source for reported 2025 response activities, 66 health centers, the reported decline and barangay clearance figures.

R6. Quezon City Government / Quezon City Health Department. (2024). Annual Procurement Plan FY2024.
https://quezoncity.gov.ph/wp-content/uploads/2024/01/Annual-Procurement-Plan-F.Y.-2024.pdf
Source for four planned vermin-abatement procurement lines; not proof of actual expenditure.

R7. Quezon City Government. (2020). Development Thrusts and Priorities / performance targets.
https://www.quezoncity.gov.ph/wp-content/uploads/2020/09/DTP-Final-v1.pdf
Source for stated dengue program targets and historical baseline, whose definitions require reconciliation.

R8. Pasig City Government. (6 March 2025). Patuloy na Pagbaba ng Kaso ng Dengue sa Pasig, Dahil sa Sama-samang Aksyon!

https://pasigcity.gov.ph/news-and-releases/patuloy-na-pagbaba-ng-kaso-ng-dengue-sa-pasig-dahil-sa-sama-samang-aksyon-465
Source for reported prevention activities and clinical coordination; the reviewed workbook does not provide a reproducible outcome series.

R9. Pasig City Government. (2025). 1st Semester Procurement Monitoring Report CY2025.

https://assets.pasigcity.gov.ph/storage/annual_procurement_plan/2025/07/13/68737fe0c9f2c17523998401st%20Semester%20Procurement%20Monitoring%20Report%20for%20CY%202025.pdf
Source for procurement-monitoring amounts whose accounting interpretation requires confirmation.

R10. Davao City Government. Ordinance No. 0401-20 and implementing rules under Executive Order No. 15, s.2023.

https://www.davaocity.gov.ph/wp-content/uploads/2023/09/Signed_RCD-2023-90847_ASD092523-17-EO-25-23_compressed.pdf
Primary legal/administrative source for the city's Mosquito-Borne Diseases Prevention and Control Program.

R11. Davao City Government. City gov't to barangays: Create Task Force vs dengue.
https://davaocity.gov.ph/health/city-govt-to-barangays-create-task-force-vs-dengue/
Implementation communication; directions and plans are not proof of completed coverage.

R12. Davao City Government. Citation tickets vs anti-mosquito borne diseases ordinance violators now ready.

https://davaocity.gov.ph/health/citation-tickets-vs-anti-mosquito-borne-diseases-ordinance-violators-now-ready/
Source for reported enforcement plans, task-force counts and Week 22 case/death comparisons; date and reporting details require validation.

R13. Davao City Government. (29 January 2026). Davao City 2025 dengue figure down vs 2024.
https://davaocity.gov.ph/health/davao-city-2025-dengue-figure-down-vs-2024/
Source for reported annual totals of 3,335 cases in 2025 and 7,157 in 2024. The comparison is not a causal evaluation.

R14. Bacolod City Government. (2025). 2025 Resolutions index; City Resolution 662 listing.
https://bacolodcity.gov.ph/2025-resolutions/
Index entry only; the full resolution and implementation records remain outstanding.

R15. SIGMA. (19 October 2026). SIGMA City Evidence and Comparison Matrix. Six-sheet workbook: Evidence Matrix, Source Register, City Comparison, Gaps and To-Verify, Data Dictionary, and LGU Name Check.

Source supplied in this conversation. It is the organizing basis for the comparative findings and explicit evidence limitations in this document.